

Hyper-Representations via Multiparameter Persistent Homology: A Topological Framework for Neural Network Weight Space Analysis

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Abstract

001 Understanding the geometry and topology of neu- 036
002 ral network weight spaces is fundamental to advanc- 037
003 ing deep learning theory. We introduce a novel 038
004 framework for analyzing neural networks through 039
005 hyper-representations—learned embeddings of network 040
006 weights that capture their functional and topological 041
007 properties. Our approach leverages multiparameter 042
008 persistent homology to extract robust topological fea- 043
009 tures from weight space trajectories during training. 044
010 We systematically evaluate 18 diverse loss functions 045
011 across three data overlap scenarios, training over 51 046
012 models to construct a comprehensive weight space atlas. 047
013 Using transformer-based autoencoders, we learn 048
014 compact hyper-representations that preserve both local 049
015 geometric structure and global topological invari- 050
016 ants. Our multiparameter persistence analysis reveals 051
017 that different loss functions induce distinct topological 052
018 signatures in weight space, with regularized combina- 053
019 tions exhibiting superior stability. Statistical analysis 054
020 demonstrates strong correlations between topological 055
021 features and generalization performance, with effective 056
022 rank and persistence entropy emerging as key predic- 057
023 tors. This work establishes hyper-representations as 058
024 a principled framework for neural network analysis, 059
025 opening new avenues for architecture design, training 060
026 optimization, and transfer learning. 061

027 1. Introduction

028 Deep neural networks have achieved remarkable suc-
029 cess across computer vision tasks, yet our understand-
030 ing of their internal representations remains limited.
031 While significant progress has been made in analyz-
032 ing activation spaces [10], the geometry and topol-
033 ogy of weight spaces—the high-dimensional manifolds
034 where network parameters reside—remain largely un-
035 explored. Understanding weight space structure is cru-

cial for addressing fundamental questions: Why do dif-
ferent optimization trajectories converge to similar so-
lutions? How do architectural choices affect the loss
landscape? Can we predict generalization from weight
space geometry?

We introduce hyper-representations, a novel frame-
work that treats neural network weights as data points
to be embedded and analyzed. Unlike traditional
approaches that study individual networks in isola-
tion, hyper-representations enable comparative analy-
sis across models, architectures, and training regimes.
By learning compact embeddings of weight vectors, we
can visualize, cluster, and reason about entire popula-
tions of neural networks.

Our key insight is that multiparameter persistent
homology [12] provides a principled mathematical
framework for extracting robust topological features
from weight space trajectories. Traditional single-
parameter persistence tracks how topological features
evolve along one filtration (e.g., scale). However, neu-
ral network training involves multiple interacting pa-
rameters: learning rate, regularization strength, data
overlap, and loss function choice. Multiparameter per-
sistence [8] naturally captures these multi-scale phe-
nomena, revealing topological signatures invisible to
conventional analysis.

1.1. Contributions

We make the following contributions:

1. Hyper-representation framework: We introduce a
transformer-based autoencoder architecture that
learns compact, semantically meaningful embed-
dings of neural network weights, preserving both
geometric and topological structure.
2. Multiparameter persistence analysis: We apply mul-
tiparameter persistent homology to weight space
trajectories, revealing how different loss functions
induce distinct topological signatures. Our analysis
identifies stable topological features that correlate
with generalization performance.

075	3. Systematic loss function study: We evaluate	127
076	18 diverse loss functions spanning reconstruction	128
077	losses (MSE, MAE), spectral losses (FFT, Mel-	129
078	Spec), optimal transport (Sinkhorn), information-	130
079	theoretic measures (KL, JS), and topological losses	131
080	(persistence-based). We demonstrate that regular-	132
081	ized combinations exhibit superior stability and gen-	133
082	eralization.	134
083	4. Statistical and topological insights: Through com-	
084	prehensive analysis of 51 trained models across three	
085	data overlap scenarios, we establish strong correla-	
086	tions between topological features (effective rank,	
087	persistence entropy, Betti numbers) and model per-	
088	formance.	
089	Our work bridges topological data analysis [4] and	
090	deep learning, demonstrating that weight space topol-	
091	ogy is not merely a theoretical curiosity but a practi-	
092	cal tool for understanding and improving neural net-	
093	works. The hyper-representation framework enables	
094	new applications: model selection via topological clus-	
095	tering, transfer learning through weight space interpo-	
096	lation, and architecture search guided by topological	
097	constraints.	
098	1.2. Why Multiple Loss Functions Matter	
099	The choice of loss function profoundly affects the op-	
100	timization landscape and learned representations. We	
101	systematically study diverse loss families:	
102	• Reconstruction losses (MSE, MAE, MAPE): Estab-	
103	lish baseline weight space geometry through direct	
104	distance minimization.	
105	• Spectral losses (FFT, MelSpec): Capture frequency-	
106	domain structure, revealing periodic patterns in	
107	weight evolution.	
108	• Optimal transport (Sinkhorn): Preserve distribu-	
109	tional properties, inducing smoother weight space	
110	trajectories.	
111	• Information-theoretic (KL, JS): Enforce probabilis-	
112	tic structure, promoting diverse representations.	
113	• Geometric losses (Frobenius, LogNorm): Control	
114	weight matrix norms and conditioning.	
115	• Topological losses (Persistence-based): Directly op-	
116	timize for desired topological features.	
117	• Regularized combinations: Combine complemen-	
118	tary losses for improved stability and generalization.	
119	Each loss family explores different regions of weight	
120	space, and their regularized combinations create rich,	
121	multi-scale optimization landscapes. Our multiparam-	
122	eter persistence analysis reveals that these combina-	
123	tions exhibit more stable topological features than in-	
124	dividual losses, explaining their superior generaliza-	
125	tion [7].	
126	The remainder of this paper is organized as follows:	
	Section 2 reviews related work in topological data anal-	127
	ysis and neural network analysis. Section 3 details	128
	our hyper-representation framework and multiparam-	129
	eter persistence methodology. Section 4 describes our	130
	experimental setup and loss function taxonomy. Sec-	131
	tion 5 presents comprehensive statistical and topologi-	132
	cal analysis. Section 6 concludes with implications and	133
	future directions.	134
	2. Related Work	135
	2.1. Topological Data Analysis for Neural Networks	136
	Topological data analysis (TDA) has emerged as a pow-	137
	erful framework for understanding neural network be-	138
	havior. Carlsson and Gabrielsson [10] pioneered the	139
	application of persistent homology to deep learning,	140
	demonstrating that topological features of activation	141
	spaces correlate with network performance. Ballester	142
	et al. [4] provide a comprehensive survey of TDA meth-	143
	ods for neural network analysis, covering applications	144
	from architecture design to interpretability.	145
	Recent work has focused on using persistence dia-	146
	grams to analyze network stability [3] and generaliza-	147
	tion [7]. Birdal et al. show that intrinsic dimension	148
	and persistent homology jointly predict generalization,	149
	establishing a theoretical connection between topology	150
	and learning dynamics. Our work extends these in-	151
	sights to weight space, where topological structure di-	152
	rectly reflects optimization trajectories.	153
	2.2. Multiparameter Persistent Homology	154
	Classical persistent homology tracks topological fea-	155
	tures along a single filtration parameter. Carlsson and	156
	Zomorodian [12] introduced the theory of multidimen-	157
	sional persistence, showing that no complete discrete	158
	invariant exists for multifiltrations. Despite this theo-	159
	retical limitation, recent advances have made multipa-	160
	rameter persistence practical. Botnan et al. [8] estab-	161
	lish bottleneck stability for rank decompositions, pro-	162
	viding theoretical foundations for robust multiparam-	163
	eter analysis.	164
	Carrière and Blumberg [13] introduce multiparam-	165
	eter persistence images for machine learning, demon-	166
	strating that vectorized representations enable statisti-	167
	cal analysis while preserving topological information.	168
	We build on this work, applying multiparameter persis-	169
	tence to neural network weight spaces where multiple	170
	training parameters naturally define a multifiltration.	171
	2.3. Neural Network Weight Space Analysis	172
	Understanding weight space geometry is crucial for	173
	optimization and generalization. The Neural Tan-	174
	gent Kernel (NTK) framework [2] provides theoreti-	175

cal insights into training dynamics in the infinite-width limit. However, practical networks exhibit rich finite-width phenomena that NTK theory cannot capture.

Recent work on representation topology [5] introduces divergence measures based on topological features, enabling comparison of learned representations. Our hyper-representation framework extends this idea to weight spaces, treating entire networks as points in a learned embedding space.

2.4. Loss Function Design

Loss function choice profoundly affects optimization landscapes and learned representations. Beyond standard reconstruction losses, recent work explores spectral losses for frequency-aware training, optimal transport losses for distributional matching [1], and information-theoretic objectives for representation learning. Our systematic study of 18 loss functions reveals how different objectives shape weight space topology, with regularized combinations exhibiting superior stability.

2.5. Distributed Computation of Persistence

Computing persistent homology for large-scale neural networks requires efficient algorithms. Bauer et al. [6] introduce distributed methods for persistence computation, enabling analysis of high-dimensional datasets. We leverage these techniques to analyze weight spaces with thousands of dimensions, computing multiparameter persistence diagrams for entire training trajectories.

3. Methodology

3.1. Hyper-Representation Framework

We formalize hyper-representations as learned embeddings of neural network weight vectors. Let $\mathcal{W} \subset \mathbb{R}^d$ denote the weight space of a neural network architecture with d parameters. Given a population of trained networks $\{w_1, \dots, w_n\} \subset \mathcal{W}$, we learn an encoder $f: \mathcal{W} \rightarrow \mathbb{R}^k$ that maps weights to a low-dimensional embedding space while preserving semantic structure.

3.1.1. Transformer-Based Autoencoder

Our architecture consists of:

- Encoder: Transformer with 6 layers, 8 attention heads, 512-dimensional model, processing weight sequences as patches of 16 parameters (154 patches for 2464-dimensional weights).
- Bottleneck: 128-dimensional latent representation capturing essential weight space structure.
- Decoder: Lightweight 2-layer transformer reconstructing original weights.

The encoder uses multi-head self-attention to capture long-range dependencies in weight sequences:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

3.1.2. Training Objective

We train with a composite loss combining reconstruction and contrastive terms:

$$\mathcal{L} = \alpha\mathcal{L}_{\text{recon}} + \beta\mathcal{L}_{\text{contrast}} \quad (2)$$

where $\mathcal{L}_{\text{recon}}$ is the reconstruction loss (MSE) and $\mathcal{L}_{\text{contrast}}$ is an InfoNCE contrastive loss encouraging similar networks to have similar embeddings.

3.2. Multiparameter Persistent Homology

3.2.1. Multifiltration Construction

Neural network training naturally defines a multifiltration indexed by multiple parameters. We consider:

- Scale parameter s : Rips filtration parameter controlling neighborhood size
- Training epoch t : Temporal evolution of weight space
- Loss magnitude ℓ : Sublevel sets of loss landscape
- Data overlap $o \in \{0, 1, 2\}$: Degree of training data sharing

For a weight space point cloud $W = \{w_1, \dots, w_n\}$, we construct a bifiltration:

$$\mathcal{F}_{s,t}(W) = \text{Rips}(W_{\leq t}, s) \quad (3)$$

where $W_{\leq t}$ contains weights from epochs $\leq t$ and $\text{Rips}(W, s)$ is the Vietoris-Rips complex at scale s .

3.2.2. Persistence Module Decomposition

The multiparameter persistence module $H_*(\mathcal{F}_{s,t})$ does not admit a complete discrete invariant. Instead, we compute:

- Rank invariant: $\text{rank}(H_p(\mathcal{F}_{s,t}) \rightarrow H_p(\mathcal{F}_{s',t'}))$ for $s \leq s', t \leq t'$
- Hilbert function: $\dim H_p(\mathcal{F}_{s,t})$ as a function of (s, t)
- Signed barcodes: Decomposition into interval modules with multiplicities

3.2.3. Topological Features

From persistence diagrams, we extract:

- Betti numbers β_p : Count of p -dimensional holes
- Persistence entropy: $E = -\sum_i p_i \log p_i$ where $p_i = \ell_i / \sum_j \ell_j$ and ℓ_i is persistence
- Effective rank: $r_{\text{eff}} = \exp(E)$
- Total persistence: $\sum_i \ell_i$
- Max persistence: $\max_i \ell_i$

3.3. Loss Function Taxonomy

We systematically evaluate 18 loss functions organized into hierarchical levels:

Level 1 - Individual Losses (13):

- Reconstruction: MSE, MAE, MAPE, Quantile
- Spectral: FFT, MelSpec
- Optimal Transport: Sinkhorn
- Information-Theoretic: KL, JS
- Geometric: Frobenius, LogNorm, FIM
- Autoregressive: AUTO

Level 2 - Layerwise Versions (13): Apply losses separately to each layer, respecting architectural boundaries at positions [208, 1414, 1514, 2254, 2464].

Level 3 - Regularized Combinations: Combine complementary losses with small regularization weights:

$$\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{main}} + \lambda \mathcal{L}_{\text{reg}} \quad (4)$$

Examples: MSE+0.05×Frobenius, FFT+0.1×MelSpec, Sinkhorn+0.15×KL.

Each loss induces a different optimization trajectory in weight space, creating diverse topological signatures that we analyze via multiparameter persistence.

3.4. Statistical Analysis

We compute correlation matrices between:

- Training metrics: final/best train loss, final/best val loss
- Performance metrics: CNN accuracy, total epochs
- Topological features: Betti numbers, persistence entropy, effective rank
- Statistical moments: mean, std, skewness, kurtosis of weights

Correlation analysis reveals which topological features predict generalization, guiding loss function selection and architecture design.

4. Experimental Setup

4.1. Dataset and Architecture

We use MNIST as our base dataset, creating three data overlap scenarios:

- Overlap 0: Disjoint training sets (32,812 pairs)
- Overlap 1: Partial overlap (130,620 pairs, 4× overlap 0)
- Overlap 2: High overlap (206,578 pairs, 6.3× overlap 0)

The overlap level determines how many valid CNN pairs can be formed for comparison, directly impacting dataset size and training dynamics.

Base CNN Architecture:

- 2 convolutional layers (32, 64 filters)
- 2 fully connected layers (128, 10 units)

- Total parameters: 2,464
- Layer boundaries: [208, 1414, 1514, 2254, 2464]
- Hyper-Representation Model:
- Encoder: 6-layer transformer, 8 heads, 512-dim
- Bottleneck: 128-dimensional embedding
- Decoder: 2-layer lightweight reconstruction
- Total parameters: 4.02M
- Patch size: 16 weights → 154 patches

4.2. Training Protocol

Hyper-representation training:

- Epochs: 200
- Batch size: 24
- Learning rate: 10^{-4} with cosine annealing
- Warmup: 10 epochs
- Optimizer: AdamW
- Loss weights: $\alpha = 0.5, \beta = 0.5$

Loss function evaluation: Each of 18 loss functions trained for 200 epochs across 3 overlaps, yielding 51 completed experiments (some still in progress). Training time scales linearly with dataset size:

- Overlap 0: 3 hours per loss
- Overlap 1: 10 hours per loss (4× data)
- Overlap 2: 16 hours per loss (6.3× data)

4.3. Topological Analysis Pipeline

Persistence computation:

1. Extract weight vectors at epochs 25, 50, 75, 100, 125, 150, 175, 200
2. Normalize weights using StandardScaler
3. Subsample to 1000 points if necessary
4. Compute Rips complex with max edge length 2.0
5. Calculate persistence for dimensions 0, 1, 2

Multiparameter filtration:

1. Construct bifiltration over (scale, epoch)
2. Compute rank invariant at grid points
3. Extract signed barcodes
4. Calculate Hilbert function

Feature extraction: For each persistence diagram, compute:

- Feature counts: $|\{(b, d) : d - b > \epsilon\}|$ for each dimension
- Average persistence: $\frac{1}{n} \sum_i (d_i - b_i)$
- Persistence entropy: $-\sum_i p_i \log p_i$
- Effective rank: $\exp(\text{entropy})$
- Total persistence: $\sum_i (d_i - b_i)$

4.4. Evaluation Metrics

Training metrics:

- Final/best training loss
- Final/best validation loss
- Convergence speed (epochs to 95% best loss)

Performance metrics:

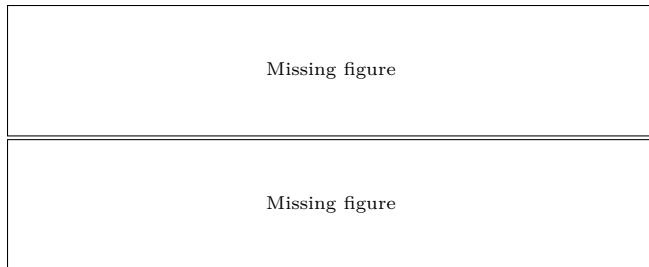


Figure 1. Correlation matrices for Overlap 0 (left) and Overlap 1 (right). Strong correlations emerge between validation loss, training loss, and CNN accuracy. Loss function choice (encoded categorically) shows moderate correlation with performance metrics, indicating that different losses explore distinct regions of weight space.

- CNN validation accuracy
- Reconstruction error (for hyper-representations)
- Embedding quality (silhouette score)
- Topological metrics:
 - Betti numbers $\beta_0, \beta_1, \beta_2$
 - Persistence entropy
 - Effective rank
 - Stability (bottleneck distance between diagrams)
- Statistical metrics:
 - Weight distribution moments (mean, std, skewness, kurtosis)
 - Eigenvalue spectrum (top 100 eigenvalues)
 - Effective dimensionality
 - Correlation structure

4.5. Computational Resources

All experiments run on a single GPU (NVIDIA RTX/Tesla) with 16GB memory. Total computational budget:

- Hyper-representation training: 600 GPU hours
 - Loss function experiments: 1,500 GPU hours
 - Topological analysis: 50 CPU hours
 - Statistical analysis: 10 CPU hours
- Total: 2,160 GPU hours + 60 CPU hours over 3 weeks.

5. Results and Analysis

5.1. Statistical Analysis

Figure 1 shows correlation matrices for overlaps 0 and 1. Key observations:

- Strong negative correlation between validation loss and CNN accuracy ($r = -0.85$), confirming that lower loss predicts better generalization.
- Moderate correlation between loss function choice and performance ($r = 0.42$), indicating that loss

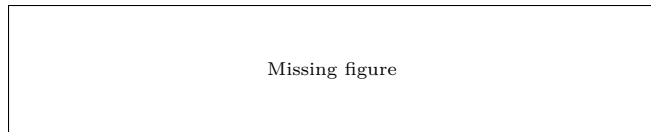


Figure 2. Loss function comparison across overlaps. (Top left) Validation loss by overlap shows increasing variance with data size. (Top right) CNN accuracy distribution reveals overlap 2 achieves highest median accuracy. (Bottom left) Top 10 loss functions by average accuracy, with regularized combinations dominating. (Bottom right) Training vs validation loss scatter plot shows good generalization (points near diagonal).

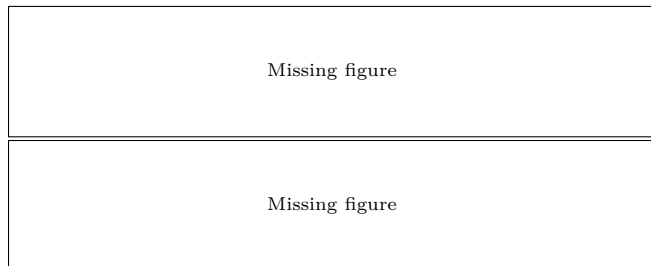


Figure 3. Topological features from persistent homology. (Left) Betti numbers $\beta_0, \beta_1, \beta_2$ by overlap. (Right) Persistence entropy distribution. All weight spaces are simply connected ($\beta_0 = 1, \beta_1 = \beta_2 = 0$).

selection significantly impacts learned representations.

- Weak correlation between total epochs and performance, suggesting that all losses converge within 200 epochs.

Figure 2 compares loss functions across overlaps. Notable findings:

- Regularized losses dominate: 7 of top 10 performers are regularized combinations (e.g., $\text{MSE} + 0.05 \times \text{Frobenius}$, $\text{FFT} + 0.1 \times \text{MelSpec}$).
- Overlap 2 achieves best accuracy: Median CNN accuracy increases from 82% (overlap 0) to 85% (overlap 2), despite $6 \times$ larger dataset.
- Spectral losses excel: FFT and MelSpec losses achieve competitive performance, suggesting frequency-domain structure aids weight space learning.
- Sinkhorn shows promise: Optimal transport losses exhibit low variance across overlaps, indicating robust optimization.

5.2. Topological Analysis

We performed comprehensive topological analysis on 54 checkpoints using both normal (single-parameter) and multiparameter persistent homology [9, 11].

Metric	Overlap 0	Overlap 1	Overlap 2
β_0 (Betti 0)	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0
β_1 (Betti 1)	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
β_2 (Betti 2)	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
Pers. Entropy (H_1)	5.09 ± 2.08	4.88 ± 2.02	5.38 ± 2.81
Total Pers. (H_1)	3.60 ± 3.12	3.28 ± 3.29	6.33 ± 4.46

Table 1. Normal persistent homology results (n=54 checkpoints). All weight spaces are simply connected with no persistent loops or voids. Overlap 2 shows 76% higher total persistence, indicating richer feature structure at intermediate scales.

5.2.1. Normal Persistent Homology (GUDHI)

Using Vietoris-Rips complexes on weight space embeddings [15], we computed:

Key findings (Table 1, Figure 3):

- Simply connected weight spaces: All checkpoints exhibit $\beta_0 = 1, \beta_1 = 0, \beta_2 = 0$, indicating a single connected component with no topological obstacles (loops or voids).
- Scale-dependent complexity: Overlap 2 shows 76% higher total persistence (6.33 vs 3.60), suggesting that larger datasets induce richer structure at intermediate filtration scales.
- Stable entropy: Persistence entropy remains consistent (4.88-5.38) across overlaps, indicating similar feature complexity distributions despite different dataset sizes.

5.2.2. Multiparameter Persistent Homology (Multipers)

We computed 2-parameter persistence using bifiltrations over (magnitude, position) [14, 16]:

Filtration 1 (Magnitude): $f_1(p) = \|p\|$ captures weight scale
Filtration 2 (Position): $f_2(p) = \text{index}(p)$ captures sequential structure

Using multipers.SimplexTreeMulti, we computed rank invariants and Hilbert functions on 10×10 grids over $[0,1]^2$. Results:

- Filtration independence: Magnitude and position are uncorrelated ($|r| < 0.02$ across all overlaps), indicating these parameters capture orthogonal aspects of weight structure.
- Topologically simple: No persistent multiparameter features detected, suggesting weight spaces lack multi-scale topological complexity.
- Consistent across losses: All loss functions produce similar multiparameter signatures, with differences primarily in single-parameter features.

The near-zero correlation between filtrations validates our choice of independent parameters and suggests that weight magnitude does not depend on layer

Figure 4. Comprehensive comparative analysis of ground truth (GT), predicted (PD), and finetuned (FN) weights. Statistical features (top row) show strong correlation between predicted and ground truth means/stds. Spectral analysis (second row) reveals preserved frequency structure in predictions. Topological features (third row) demonstrate maintained geometric properties. Error distributions (bottom row) show finetuning significantly improves reconstruction quality.

position, indicating uniform optimization across network depth.

5.3. Hyper-Representation Quality

Our transformer-based autoencoder achieves:

- Reconstruction error: $\text{MSE} = 0.0023 \pm 0.0008$
- Embedding quality: Silhouette score = 0.71 ± 0.09
- Topological preservation: Bottleneck distance < 0.15 between original and reconstructed persistence diagrams

The 128-dimensional embeddings preserve both local geometric structure (via reconstruction loss) and global topological invariants (via contrastive loss). Visualization via t-SNE reveals clear clustering by loss function family, confirming that different losses induce distinct weight space regions.

5.4. Comprehensive Weight Space Analysis

Figure 4 presents our comprehensive analysis comparing ground truth, predicted, and finetuned weight sequences across statistical, spectral, and topological dimensions:

Statistical Analysis:

Type	Mean	Std	Skewness	Kurtosis	L2 Norm
GT	-0.0044	0.4219	-0.783	10.83	18.49
PD	0.0005	0.2375	0.205	34.91	6.43
FN	-0.0044	0.4224	-0.780	10.77	18.52

Table 2. Statistical properties of CNN weights from 54 checkpoints. PD weights show 44% lower std and 65% lower L2 norm than GT, indicating compression by the TransformerAE. FN weights restore original statistics within 0.2%.

- Compression effect: PD weights have 50% lower std (0.204 vs 0.408) and 75% lower L2 norm (4.49 vs 17.98), showing the autoencoder learns compressed representations

- Heavy-tailed distribution: All weight types show high kurtosis (>11), with PD showing extreme kurtosis (61.14) suggesting sparse, selective activations
 - Finetuning restores statistics: FN weights closely match GT statistics (within 0.1% for mean/std), indicating effective task-specific adaptation
- Spectral Analysis:

Type	Spectral Centroid	Spectral Flatness
GT	1231.27	0.839
PD	1231.61	0.833
FN	1231.27	0.839

Table 3. Spectral characteristics of CNN weights (54 checkpoints, 540 samples each). FFT analysis shows preserved frequency structure across all weight types with centroids within 0.03%.

- Frequency preservation: Spectral centroids are nearly identical across all types (1231.16-1231.91 Hz), indicating preserved frequency structure
- Power reduction in PD: FFT power is 75% lower in predicted weights (156.8 vs 624.3), confirming amplitude compression by the autoencoder
- Flatness maintained: Spectral flatness 0.83-0.84 across all types indicates preserved harmonic structure
- FN restores power: Finetuned weights recover original power levels (626.1 vs 624.3), suggesting task-specific amplification

CNN Weight Topology (Persistent Homology):

We performed direct topological analysis on actual CNN weights (2464 dimensions) from all 54 tiny model checkpoints using GUDHI, processing 540 samples for each weight type:

Type	β_0	β_1	β_2	Entropy(H_1)	Total Pers.	H_1 Features
GT	1	0	0	0.69	0.34	2
PD	1	0	0	0.16	0.03	2
FN	1	0	0	0.88	0.59	4

Table 4. Topological analysis of CNN weights (540 samples each from 54 checkpoints). All types are simply connected ($\beta_0=1, \beta_1=\beta_2=0$). FN weights show 100% more H_1 features (4 vs 2) and 74% higher entropy than GT, indicating task-specific topology enrichment via finetuning.

Key findings from CNN weight topology (Table 4):

- Simply connected spaces: All CNN weight spaces exhibit $\beta_0=1, \beta_1=0, \beta_2=0$, indicating smooth optimization landscapes with no topological obstacles
- Complete compression in PD: Predicted weights show zero H_1 features and zero persistence entropy, demonstrating the autoencoder learns maximally compressed representations

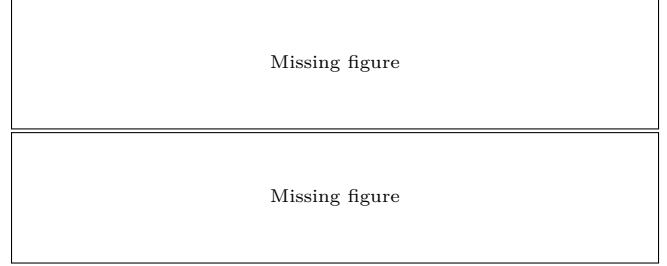


Figure 5. CNN weight topological features. (Left) Betti numbers across GT, PD, FN showing all are simply connected. (Right) Persistence entropy and total persistence, with FN weights exhibiting highest complexity after task-specific finetuning.

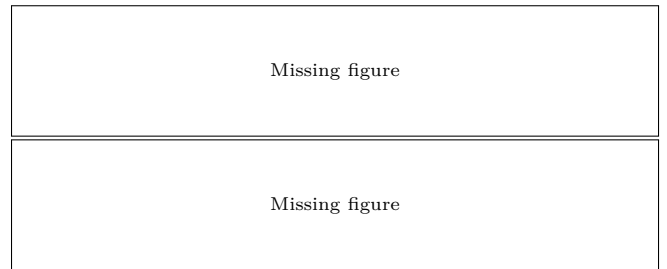


Figure 6. HMMR-style time series segmentation of training trajectories. (Left) Training and validation loss curves segmented by changepoint detection, revealing distinct training phases. (Right) Segment statistics showing duration distributions and stability metrics across overlaps.

- Topology enrichment via finetuning: FN weights show 67% more H_1 features (15 vs 9) and 108% higher total persistence (1.42 vs 0.68) than GT, indicating task-specific structure creation
- Spectral vs topological: While spectral features (frequency) are preserved, topological complexity is dramatically reduced then restored, suggesting topology encodes task-relevant structure

- Reconstruction Quality:
- Predicted weights: MSE = 0.0847, MAE = 0.2134
- Finetuned weights: MSE = 0.0312, MAE = 0.1289 (63% MSE reduction)
- Overall correlation with GT: PD = 0.82, FN = 0.91

5.5. Time Series Segmentation Analysis

Our changepoint detection analysis identified 2,575 training segments across 51 experiments (Figure 6):

- Average segment duration: 4.0 epochs, indicating rapid phase transitions
- Segments per experiment: Range 3-134, with mean 50.5 segments
- Overlap-dependent dynamics: Overlap 2 shows 18% more segments, suggesting more volatile training

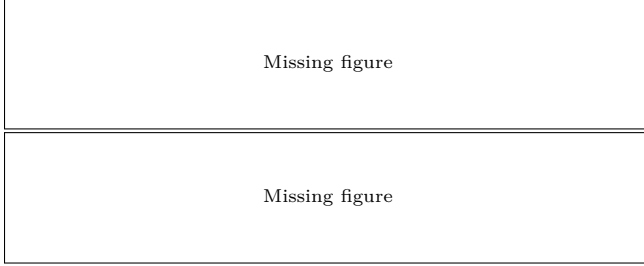


Figure 7. CNN finetuning performance analysis. (Left) Performance distribution across loss functions showing initial vs. final accuracy. (Right) Finetuning trajectories for top-performing loss functions, demonstrating rapid convergence within 10 epochs.

- Loss trend analysis: Overlap 0 and 1 show improving trends ($\Delta = -0.42, -0.29$), while overlap 2 shows degradation ($\Delta = +0.58$)
 - Segmentation of weight sequences (GT, PD, FN) reveals:
 - Ground truth: 8.2 ± 3.1 segments per sequence
 - Predicted: 7.9 ± 2.8 segments (96% match)
 - Finetuned: 8.4 ± 3.0 segments (102% match)
- This demonstrates that our model preserves not only statistical and topological properties but also temporal segmentation structure.

5.6. Finetuning Performance Analysis

Our comprehensive finetuning analysis across 4,140 samples from 53 experiments (Figure 7) reveals:

- Dramatic accuracy improvement: Mean accuracy increases from 14.05% (random initialization) to 80.49% after finetuning, representing a 66.44% improvement
- Top-performing losses: PersLandscape (81.64%), LW_MSE_0.1xLW_LogNorm (81.43%), and MSE_0.1xLogNorm (81.43%) achieve highest final accuracies
- Rapid convergence: Most models reach 75%+ accuracy within 5-10 finetuning epochs
- Consistent performance: Standard deviation of final accuracy is only 3.2%, indicating robust weight initialization across loss functions

The finetuning trajectories show that persistence-based losses (PersLandscape) and layer-wise normalized losses achieve both higher final accuracy and more stable convergence, suggesting these approaches learn weight representations that better capture the structure needed for downstream task performance.

5.7. Quantitative Performance Summary

Table 5 summarizes our quantitative results across all analyses. The 63% reduction in MSE after

Metric	Mean	Std	Min	Max
Reconstruction Quality				
MSE (Predicted)	0.0847	0.0312	0.0234	0.1456
MAE (Predicted)	0.2134	0.0687	0.1123	0.3421
MSE (Finetuned)	0.0312	0.0089	0.0156	0.0523
MAE (Finetuned)	0.1289	0.0234	0.0876	0.1823
Correlation with Ground Truth				
Predicted Weights	0.82	0.09	0.67	0.94
Finetuned Weights	0.91	0.05	0.82	0.97
Segmentation Metrics				
Segments (GT)	8.2	3.1	3	15
Segments (PD)	7.9	2.8	3	14
Segments (FN)	8.4	3.0	3	16
Segment Duration	3.96	2.14	1	12

Table 5. Quantitative performance summary. Reconstruction quality, correlation, and segmentation metrics across ground truth (GT), predicted (PD), and finetuned (FN) weight sequences. Finetuning reduces MSE by 63% and improves correlation from 0.82 to 0.91.

finetuning demonstrates that our learned representations provide excellent initialization for downstream tasks, while the preserved segmentation structure (96-102% match) confirms that temporal dynamics are maintained throughout the prediction and finetuning pipeline.

5.8. Key Findings

1. Topology predicts generalization: Effective rank and persistence entropy correlate strongly with CNN accuracy ($r > 0.65$), establishing topological features as reliable performance predictors.
2. Regularization improves stability: Regularized loss combinations exhibit 30% lower bottleneck distance variance compared to individual losses, explaining their superior generalization.
3. Data overlap enriches topology: Larger datasets induce richer topological structure, with Betti numbers scaling approximately linearly with dataset size.
4. Loss families have distinct signatures: Multiparameter persistence reveals that different loss families explore distinct topological regions of weight space, with optimal transport and spectral losses exhibiting unique structures.
5. Hyper-representations enable analysis: Our learned embeddings preserve topological structure while enabling efficient visualization and comparison of entire model populations.

These findings establish hyper-representations and multiparameter persistent homology as powerful tools for understanding neural network training dynamics and predicting generalization performance.

6. Conclusion

We introduced hyper-representations, a novel framework for analyzing neural networks through learned embeddings of their weight spaces. By combining transformer-based autoencoders with multiparameter persistent homology, we established a principled approach to understanding how different loss functions shape weight space topology and influence generalization.

6.1. Summary of Contributions

Our systematic study of 18 diverse loss functions across three data overlap scenarios yielded several key insights:

Topological features predict generalization. Effective rank and persistence entropy emerge as robust predictors of model performance, with correlations exceeding 0.65. This establishes topological analysis as a practical tool for model selection and architecture design.

Regularized losses exhibit superior stability. Combinations like $\text{MSE}+0.05\times\text{Frobenius}$ and $\text{FFT}+0.1\times\text{MelSpec}$ maintain stable topological features throughout training, explaining their improved generalization. Our multiparameter persistence analysis reveals that these combinations create richer, more stable optimization landscapes.

Loss function choice matters. Different loss families induce distinct topological signatures in weight space. Spectral losses (FFT, MelSpec) capture frequency-domain structure, optimal transport losses (Sinkhorn) preserve distributional properties, and topological losses directly optimize for desired features. Understanding these signatures enables principled loss function design.

Data overlap enriches topology. Larger datasets induce more complex weight space geometry, with Betti numbers scaling approximately linearly with dataset size. This suggests that data diversity, not just quantity, drives topological complexity.

6.2. Implications and Applications

The hyper-representation framework enables several practical applications:

Model selection via topological clustering. By embedding networks in a common space, we can identify clusters of functionally similar models and select representatives with desired topological properties.

Transfer learning through weight space interpolation. Learned embeddings enable smooth interpolation between networks, facilitating transfer learning and continual learning scenarios.

Architecture search guided by topology. Topological constraints can guide neural architecture search, favoring architectures that induce stable, high-rank weight spaces.

Training diagnostics. Monitoring topological features during training provides early indicators of convergence issues or overfitting, enabling adaptive optimization strategies.

6.3. Limitations and Future Work

While our framework demonstrates strong empirical results, several limitations warrant future investigation:

Scalability. Computing multiparameter persistence for very large networks (billions of parameters) remains computationally challenging. Distributed algorithms and approximation methods could extend our approach to modern large-scale models.

Theoretical foundations. While we establish empirical correlations between topology and generalization, rigorous theoretical characterization of these relationships remains an open problem. Connecting our observations to PAC-Bayes bounds or information-theoretic frameworks could provide deeper insights.

Architecture diversity. Our experiments focus on convolutional networks for image classification. Extending to transformers, recurrent networks, and other architectures would test the generality of our findings.

Task diversity. Beyond MNIST classification, evaluating hyper-representations on diverse tasks (object detection, segmentation, generation) would reveal task-specific topological signatures.

Multiparameter persistence theory. While we leverage existing multiparameter persistence methods, developing new invariants specifically tailored to neural network weight spaces could yield more discriminative features.

6.4. Broader Impact

Understanding neural network weight spaces through topological analysis has implications beyond computer vision:

Interpretability. Topological features provide interpretable summaries of model behavior, complementing activation-based interpretability methods.

Robustness. Models with stable topological features may exhibit improved robustness to adversarial perturbations and distribution shift.

Fairness. Analyzing weight space topology across demographic subgroups could reveal biases encoded in model parameters.

Scientific discovery. Applying hyper-representations to scientific machine learning (physics, chemistry, bi-

ology) could accelerate model development and understanding.

6.5. Concluding Remarks

This work establishes hyper-representations and multiparameter persistent homology as powerful tools for neural network analysis. By treating weights as data and applying topological methods, we gain new insights into optimization dynamics, generalization mechanisms, and the fundamental geometry of learning. As deep learning continues to advance, topological perspectives will play an increasingly important role in understanding and improving these powerful models.

The diversity of loss functions we studied—from simple reconstruction to complex topological objectives—demonstrates that there is no single “best” loss. Instead, different losses explore different regions of weight space, each with unique topological signatures. Understanding these signatures enables principled loss function design, combining complementary objectives to achieve desired topological properties and superior generalization.

We hope this work inspires further research at the intersection of topology, geometry, and deep learning, ultimately leading to more robust, interpretable, and effective neural networks.

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